

Anish Acharya, PhD

Senior Research Scientist
Amazon Web Services (AWS)

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Website • Scholar • GitHub • LinkedIn

Education

- **The University of Texas at Austin** Austin, TX
PhD, Electrical and Computer Engineering Fall, 2019 – Spring, 2025
Advisor(s): **Prof. Inderjit S. Dhillon** and **Prof. Sujay Sanghavi**
PhD Committee: **Prof. Alex Dimakis**, **Prof. Ludwig Schmidt**, and **Prof. Nhat Ho**
PhD Thesis: **Robust and Efficient Learning in High Dimensions from Noisy Data**
- **University of California, Irvine** Irvine, CA
M.S., Electrical Engineering and Computer Science Fall, 2013 – Fall, 2014
- **Jadavpur University** Kolkata, India
B.E., Electronics and Instrumentation Engineering Spring, 2009 – Spring, 2013
Advisor(s): **Amitava Gupta** and **Shantanu Das**

Research Interests

My recent focus is **Recursive Self Improvement (RSI)**: the loop in which an agentic system proposes changes to its own policy (prompts, tools, orchestration logic and weights), scores them, and keeps what scored better. I work on both halves of that loop, the **signal** it optimizes against and the **search** that exploits it, together with the theory of when iterating provably helps. More broadly: **robust reward modelling and evaluation** under biased or adversarial supervision, **multi-agent systems**, **constrained combinatorial search**, and **robust, distributed and federated optimization**.

Research Experience

- **Amazon** Austin, TX
Senior Research Scientist, AWS (Full Time) Spring, 2025 – Present
Building Decentralized AI Marketplace.
Lead the full life-cycle of AWS **multi-agent orchestration meshes** – architecture, core research, production integration, and eval-driven iteration. Research on **recursive self improvement** – robust reward modeling (**RoPoLL**), prompt optimization, and orchestration search (**OrchOpt**) – directly powers: **AI-assisted Product Listing**, AWS Partner Assistant, AWS Summit NY 2026 (**blog**, **docs**, **video**); **AWS Partner Central Agents** – partner onboarding and funding, served over MCP, 2026 (**onboarding**, **funding**, **MCP**); **AWS Marketplace Agent Mode**, re:Invent 2025 (**blog**, **product**, **ZDNet**, **CRN**, **The Ravit Show**, **Inside Partnering**).
- **University of Texas at Austin** Austin, TX
Graduate Research Assistant (Part Time) Fall, 2019 – Spring, 2025
Doctoral research on **robust, sample-efficient learning in high dimensions**: robust estimation under label noise and Byzantine corruption, distributed and federated optimization, and data-efficient subset selection (**thesis**). Affiliated with the **Institute for Foundations of Machine Learning (IFML)**.
- **Eidon AI** Austin, TX
Head of AI Research (Part Time) Summer, 2024 – Winter, 2025
The AI is the Blockchain, the Blockchain is the AI.
Architected and launched a **multi-agent dynamic point-based system** integrating blockchain with AI: evaluation metrics for content quality, AI alignment, and community standards, and a **tokenomics framework for reward distribution** across the decentralized network (**Technical Docs**).
- **Meta** Menlo Park, CA
Student Researcher (Part Time) Fall, 2022
Research Scientist (Intern) Summer, 2022
Student Researcher (Part Time) Fall, 2021
Research Scientist (Intern) Summer, 2021
With Meta Capacity Engineering and Meta Fundamental AI Research (FAIR), hosted by **Li Jing**, **Bhargav Bhushanam**, **Dhruv Choudhary**, and **Michael Rabbat**: online, low-resource training to scale Facebook’s deep recommendation models. Developed **positive-unlabeled contrastive representation learning** for weakly supervised click-through-rate prediction, learning from positives and unlabeled data without negative sampling – **ICML 2022 PODS workshop** (**paper**) and **extended version**.
- **Microsoft** Redmond, WA
Research Scientist (Intern) Summer, 2020
As part of the **Speech and Dialog Research Group** (hosted by **Kenichi Kumatani** and **Dimitrios Dimitriadis**), developed algorithms

for training large-scale models robust to **adversarial, Byzantine, and noisy client updates** in the federated setting; built **OpenMSFTL**, an open-source federated learning platform ([github](#)).

- **Amazon** **Sunnyvale, CA**
Applied Scientist, AGI (Full Time) *Summer, 2018 – Spring, 2019*
Applied Scientist, Alexa AI (Full Time) *Spring, 2018 – Summer, 2018*
Applied Scientist, Alexa ML (Full Time) *Winter, 2017 – Spring, 2018*
Software Development Engineer, Amazon Devices (Full Time) *Summer, 2016 – Winter, 2017*
Shipped NLP domain models shared across Alexa devices – Timers, Alarms, Reminders, Calendar, Communications – on Echo Show, Fire TV, and other Alexa IoT devices. **Primary inventor on: Alexa Conversations**, goal-oriented dialog generation from dialog templates, APIs, and entity data ([paper](#), [patent](#), [skill](#), [Amazon Science](#), [CNN](#), [Forbes](#), [VentureBeat](#)); **online embedding compression for deployment-constrained NLU** – low-rank factorization folded into training, so task data shapes the compression itself, deployed across a large fleet of per-skill models ([paper](#), [patent](#), [Amazon Science](#), [Amazon Blog](#), [Packt](#), [Hacker News](#)).
- **eBay Inc.** **San Jose, CA**
Data Scientist, Customer Insights and Analytics (Contract – Manthan Systems Inc.) *Fall, 2015 – Summer, 2016*
Developed **deep recommendation algorithms** to personalize buyer search and targeted advertising, e.g., **generative models for personalized emails**.
- **Schlumberger** **Houston, TX**
Data Scientist (Full Time) *Spring, 2015 – Fall, 2015*
Early member of the **Digital Transformation Services** AI research group: **failure prediction** for deployed oil service instruments and **oil level estimation** from extraction life-cycle sensor data.
- **FEM Inc.** (acquired by **Gracenote**, a Nielsen company) **Los Angeles, CA**
Data Engineer (Full Time) *Winter, 2015*
With the co-founders: NLP models for a content recommendation system, and visualization tooling for an interactive analytics dashboard.
- **Toyota Technological Institute Chicago (TTIC)** **Chicago, IL; Ann Arbor, MI**
Visiting Research Scientist (Intern) *Summer, 2014*
Hosted by **Prof. David McAllester**: real-time stereo algorithm for 3D scene reconstruction for autonomous vehicles, plus a stereo evaluation benchmark based on a slanted-plane scene model; built for **Toyota Research in North America (TRINA)**.

Teaching Experience

- **Amazon Web Services** **Austin, TX**
Instructor
AWS GenAI Training: **Embedding AI into Products: Evaluation-Driven Development** *Summer, 2026*
AWS GenAI Training: **Embedding AI into Products: Governance and Guardrails** *Spring, 2026*
- **University of Texas at Austin** **Austin, TX**
Instructor *Fall, 2025*
Directed Reading Program (DiRP): **Natural Language Optimization (NLO) for LLMs**
Teaching Assistant
MIS 285N: **Introduction to Deep Learning**, Instructors: Prof. C. Caramanis, Prof. A. Dimakis *Spring, 2024*
MIS 185N: **Emerging Trends in Machine Learning**, Instructor: Prof. S. Shakkottai *Spring, 2024*
ECE 461P: **Data Science Principles**, Instructor: Prof. S. Sanghavi *Fall, 2023*
- **University of California Irvine** **Irvine, CA**
Teaching Assistant
EECS 50: **Discrete Time Signals and Systems** (Undergraduate) *Fall, 2014*
EECS 280A: **Advanced Engineering Electromagnetics I** (Graduate) *Spring, 2014*
EECS 118: **Introduction to Knowledge Management and Software Engineering** (Undergraduate) *Winter, 2014*
EECS 119: **Introduction to VLSI** (Undergraduate) *Fall, 2013*

Professional Service

- **Reviewer, Conferences:**
 - **International Conference on Machine Learning (ICML)**
 - **Conference on Neural Information Processing Systems (NeurIPS)**
 - **International Conference on Learning Representations (ICLR)**
 - **International Conference on Artificial Intelligence and Statistics (AISTATS)**
 - **Conference on Uncertainty in Artificial Intelligence (UAI)**
 - **Annual Meeting of the Association for Computational Linguistics (ACL)**
 - **IEEE International Symposium on Information Theory (ISIT)**

- **Reviewer, Journals:**
 - [Journal of Machine Learning Research](#) (MIT Press)
 - [Transactions on Knowledge and Data Engineering](#) (IEEE)
 - [Pattern Recognition Letters](#) (Elsevier)
 - [Cognitive Neurodynamics](#) (Springer Nature)
 - [Discover Electronics](#) (Springer Nature)
- **Program Committee:**
 - [International Conference on Machine Learning \(ICML\)](#)
 - [AAAI Conference on Artificial Intelligence \(AAAI\)](#)

Honors and Awards

- **Silver Reviewer** award at ICML 2026. Spring, 2026.
 - **Travel Grants**, UT Austin – ML conferences (NeurIPS, ICML, AISTATS, UAI) and Simons Institute, UC Berkeley. 2021 – 2024.
 - **Meta PhD Research Grant** of \$100,000 Spring, 2021 – Spring, 2023.
 - **Meta PhD Research Fellowship Finalist** – AI HW/SW co-design ([Press Release](#)) Spring, 2022.
 - Awarded Bachelor of Engineering with **Highest Honors** (First Class), Jadavpur University Fall, 2013
 - **INSPIRE Fellowship** and **MHRD Fellowship for Academic Excellence**, Govt. of India. Fall, 2009 – Spring, 2013.
 - **State Rank 5** of ~700,000 candidates, West Bengal Board of Secondary Education ([The Telegraph](#)). Spring, 2007
 - **Mamraj Agarwal Rashtriya Puraskar** (National Award for Academic Excellence), from the **Governor of West Bengal**. Summer, 2007
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Granted US Patents (Primary Inventor)

1. (Amazon) **Goal Oriented Dialog Generation using Dialog Template, API and Entity data**; July, 2022. [USPTO](#)
A. Acharya, A. Metallinou, T. Chung, S. Paul, S. Chandra, C. Lin, D. H. Tur, A. Mandal
 2. (Amazon) **Natural Language Processing: Amazon Alexa AI**; December, 2020. [USPTO](#)
A. Acharya, A. Metallinou, R. Goel, I. Dhillon
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Publications

1. **SPLICE: Structured Prompt Local Iterative Combinatorial Evolution.**
A. Acharya, P. Studans, A. Dhanda, N. V. Rao, V. A. Khatri, S. A. Niaki, B. Verkhovsky.
Conference on Neural Information Processing Systems (NeurIPS), 2026
2. **RoPoLL: Robust Panel of LLM Judges.**
A. Acharya, K. Pan, B. Verkhovsky.
International Conference on Machine Learning (ICML), 2026
Workshop on Statistical Frameworks for Uncertainty in Agentic Systems (AgenticUQ)
• [OpenReview](#) • [Arxiv](#) • [Amazon Science](#)
3. **OrchOpt: Validation-Gated Architecture Search for Self-Organizing Multi-Agent Orchestration.**
A. Dhanda, S. A. Niaki, A. Acharya, B. Verkhovsky.
Conference on Language Modeling (COLM), 2026
2nd Workshop on Lifelong Agents: Learning, Aligning, and Evolving
• [OpenReview](#)
4. **Geometric Median Matching for Robust k-Subset Selection from Noisy Data.**
A. Acharya, S. Sanghavi, A. G. Dimakis, I. S. Dhillon.
International Conference on Machine Learning (ICML), 2025
• [PMLR](#) • [OpenReview](#) • [Arxiv](#)
5. **Neural Distributed Source Coding.**
A. Nagle, J. Whang, A. Acharya, H. Kim, A. Dimakis.
IEEE Journal on Selected Areas in Information Theory, 2024
• [IEEE Xplore](#) • [Arxiv](#)
6. **Faster Non-Convex Federated Learning via Global and Local Momentum.**
R. Das, A. Acharya, A. Hashemi, S. Sanghavi, I. Dhillon, U. Topcu.
The Conference on Uncertainty in Artificial Intelligence (UAI), 2022
• [OpenReview](#)

SPOTLIGHT

7. **Robust Training in High Dimensions via Block Coordinate Geometric Median Descent.**
A. Acharya, A. Hashemi, P. Jain, S. Sanghavi, I. Dhillon, U. Topcu.
International Conference on Artificial Intelligence and Statistics (AISTATS), 2022
 • [PMLR](#)
8. **Positive Unlabeled Contrastive Learning.**
A. Acharya, S. Sanghavi, L. Jing, B. Bhushanam, D. Choudhary, M. Rabbat, I. Dhillon.
International Conference on Machine Learning (ICML), 2022
 Principles of Distribution Shift (PODS) Workshop
 • [Proceedings](#) • [Arxiv](#)
9. **LDKP: A Dataset for Identifying Keyphrases from Long Scientific Documents.**
 D. Mahata, N. Agarwal, D. Gautam, A. Kumar, S. Parekh, Y. K. Singla, **A. Acharya**, R. R. Shah.
ACM International Conference on Information and Knowledge Management (CIKM), 2022
 DL4SR'22: Workshop on Deep Learning for Search and Recommendation
 • [CEUR-WS](#) • [Arxiv](#) • [HF Dataset](#)
10. **Alexa Conversations: An Extensible Data-driven Approach for Building Task-oriented Dialogue Systems.** DEMO
A. Acharya, S. Adhikari, S. Agarwal, V. Auvray, N. Belgamwar, A. Biswas, S. Chandra, T. Chung, M. Zarandi, R. Gabriel, S. Gao, R. Goel, D. Hakkani-Tur, J. Jezabek, A. Jha, J. Kao, P. Krishnan, P. Ku, A. Goyal, C. Lin, Q. Liu, A. Mandal, A. Metallinou, V. Naik, Y. Pan, S. Paul, V. Perera, A. Sethi, M. Shen, N. Strom and E. Wang.
Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2021
 • [ACLAnthology](#)
11. **GupShup: An Annotated Corpus for Abstractive Summarization of Open-Domain Code-Switched Conversations.**
 L. Mehnaz, D. Mahata, A. Kumar, U. S. Gunturi, R. Jain, G. Gupta, R. Gosangi, I. G. Lee, **A. Acharya**, R. Shah.
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2021
 • [ACLAnthology](#)
12. **On the Benefits of Multiple Gossip Steps in Communication Constrained Federated Learning.**
 A. Hashemi, **A. Acharya**, R. Das, H. Vikalo, S. Sanghavi, I. Dhillon.
IEEE Transactions on Parallel and Distributed Systems, 2021
 • [IEEE Xplore](#)
13. **Online Embedding Compression for Text Classification using Low Rank Matrix Factorization.** SPOTLIGHT
A. Acharya, R. Goel, A. Metallinou, I. Dhillon.
AAAI Conference on Artificial Intelligence (AAAI), 2019
 • [Proceedings](#)
14. **Extending The Concept of Analog Butterworth Filter For Fractional Domain.**
A. Acharya, S. Das, I. Pan, S. Das.
Signal Processing (Elsevier), 2014
 • [ScienceDirect](#) • [Arxiv](#)
15. **Simulation studies on the design of optimum PID controllers to suppress chaotic oscillations in a family of Lorenz-like multi-wing attractors.**
 S. Das, **A. Acharya**, I. Pan.
Mathematics and Computers in Simulation (Elsevier), 2014
 • [ScienceDirect](#) • [Arxiv](#)
16. **Stability Analysis Of Delayed System Using Bode's Integral.** ORAL
A. Acharya, D. Mitra, K. Halder.
International Conference on Computer Communication and Informatics (ICCCI), 2013
 • [IEEE Xplore](#)
17. **Optimum PID Control of Multi-wing Attractors in A Family of Lorenz-like Chaotic Systems.** ORAL
A. Acharya, S. Das, I. Pan.
International Conference on Computing, Communication and Networking Technologies (ICCCNT), 2012
 • [IEEE Xplore](#)
18. **Optimized Quality Factor of Fractional Order Analog Filters with Band-Pass and Band-Stop Characteristics.** ORAL
 A. Pakhira, S. Das, **A. Acharya**, I. Pan, S. Saha.
International Conference on Computing, Communication and Networking Technologies (ICCCNT), 2012
 • [IEEE Xplore](#)

19. **Identification of nonlinear systems from the knowledge around different operating conditions: a feed-forward multilayer Artificial Neural Net (ANN) based approach.** ORAL
S. Saha, S. Das, **A. Acharya**, A. Kumar, S. Mukherjee, I. Pan, A. Gupta.
International Conference on Parallel, Distributed and Grid Computing (PDGC), 2012
• [IEEE Xplore](#)
20. **Least square and instrumental variable system identification of AC servo position control system with fractional Gaussian noise.** ORAL
S. Das, A. Kumar, I. Pan, **A. Acharya**, S. Das, A. Gupta.
International Conference on Energy, Automation and Signal (ICEAS), 2011
• [IEEE Xplore](#)
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Invited Talks and Presentations

- Fall, 2026. (Seattle, WA) **Amazon Machine Learning Conference (AMLC)** – Prompt Optimization (Oral)
 - Fall, 2026. (Seattle, WA) **Amazon Machine Learning Conference (AMLC)** – Robust LLM Jury (Oral)
 - Summer, 2026. (Seoul, South Korea) **ICML 2026** – Robust LLM Jury (Poster)
 - Summer, 2026. (New York City, NY) **AWS Marketplace Science** – Multi-Agent Optimization (Invited Talk)
 - Summer, 2026. (Virtual) **AI Across AWS** – Agent Optimization (Invited Talk)
 - Spring, 2026. (Virtual) **AWS Town Hall** – Prompt Optimization (Invited Talk)
 - Fall, 2025. (Seattle, WA) **Amazon Fulfillment Simulation Symposium (AFSS)** – Robust Sampling (Invited Talk)
 - Summer, 2025. (Vancouver, Canada) **ICML 2025** – Robust Sampling (Recorded Talk, Poster)
 - Fall, 2023. (Austin, TX) **Future of Chatbots and Conversational AI Summit (FOCA 2023)** (Keynote, Panel)
 - Fall, 2022. (Virtual) **Meta: Fundamental AI Research (FAIR)** (Invited Talk)
 - Fall, 2022. (Berkeley, CA) **Simons Institute for the Theory of Computing** – Robust Optimization (Poster)
 - Fall, 2022. (Virtual) **Meta: Ads ML Research** (Invited Talk)
 - Summer, 2022. (Eindhoven, Netherlands) **UAI 2022** – Faster Non-Convex Federated Learning (Poster)
 - Summer, 2022. (Baltimore, MD) **ICML 2022** – Robust Representation Learning (Recorded Talk, Poster)
 - Spring, 2022. (Austin, TX) **6g@UT Forum** – Robust Federated Learning (Poster)
 - Spring, 2022. (Virtual) **AISTATS 2022** – Robust Federated Learning (Recorded Talk, Poster)
 - Spring, 2022. (Virtual) **University of Exeter – Department of Mathematics** – Robust Federated Learning (Invited Talk)
 - Fall, 2021. (Virtual) **University of Denver – Department of Computer Science** – Robust Federated Learning (Guest Lecture)
 - Fall, 2021. (Berkeley, CA) **Simons Institute for the Theory of Computing** – Robust Federated Learning (Poster)
 - Fall, 2021. (Virtual) **Meta: Fundamental AI Research (FAIR)** – Robust Representation Learning (Invited Talk)
 - Summer, 2021. (Virtual) **Meta: Capacity Engineering Research** – Robust Representation Learning (Invited Talk)
 - Spring, 2021. (Virtual) **NSF TRIPODS** – Efficient Decentralized Optimization (Poster)
 - Summer, 2020. (Virtual) **Microsoft: Azure AI** – Robust Optimization (Invited Talk)
 - Fall, 2019. (Austin, TX) **UT Austin Graduate Research Showcase** – Efficient ML (Poster)
 - Fall, 2019. (Austin, TX) **Texas Wireless Summit** – Efficient ML (Poster)
 - Spring, 2019. (Honolulu, HI) **AAAI 2019** – Efficient ML (Oral)
 - Summer, 2018. (Seattle, WA) **Amazon Machine Learning Conference (AMLC)** – Efficient ML (Oral)
 - Winter, 2013. (Coimbatore, India) **IEEE ICCCI 2013** – Stability of Delayed Systems (Oral)
 - Fall, 2012. (Solun, India) **IEEE PDGC 2012** – Neural Network System Identification (Oral)
 - Summer, 2012. (Coimbatore, India) **IEEE ICCCNT 2012** – Chaos Control via Optimal PID (Oral)
 - Summer, 2012. (Coimbatore, India) **IEEE ICCCNT 2012** – Fractional Order Analog Filters (Oral)
 - Winter, 2011. (Bhubaneswar, India) **IEEE ICEAS 2011** – System Identification under Fractional Gaussian Noise (Oral)
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